

House Price Prediction Using Machine Learning

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Dataset: Housing.csv

Project: House Price Prediction

Project Overview:

The aim of this research project was to explore housing data and create machine learning models that were able to predict house prices according to different parameters. This data set comprised 545 observations and 13 features such as house area, number of bedrooms, bathrooms, stories, parking, air conditioning, and furnishings.

Data Preparation:

The Dataset is then loaded with the help of Pandas and analyzed to have some idea about its structure. Any missing value or duplicates present in the dataset were examined but no problem of data quality was observed. Categorical data such as mainroad, guestroom, basement, air conditioning, and furniture status have been numerically encoded.

Model Development:

Data was split into train and test datasets using an 80:20 ratio. The two models that have been trained and evaluated include:

- Linear Regression
- Random Forest Regressor

Performance measures for these models included MAE, RMSE, and R2 Score.

Model Results:

Metric	Linear Regression	Random Forest
MAE	970,043	1,049,866
RMSE	1,324,507	1,418,191
R2 Score	0.6529	0.6021

From the above observations, it can be concluded that Linear Regression is superior to Random Forest Regressor in this particular case, with lesser prediction errors and a high value of R2.

Visual Analysis:

There were several types of visualizations that were generated:

- Distribution of house prices using a histogram.
- Correlation heatmap to identify relationships between features and price.
- Actual vs. Predicted price scatter plot to assess model performance.

Analysis revealed that the feature area, bathrooms, number of stories, parking, and air conditioning has significant impact on the prices of houses.

Conclusion:

The Linear Regression model managed to explain about 65% of the variability of housing price and generated relatively accurate predictions. One interesting discovery made during the analysis is that Linear Regression performed better than the Random Forest Regressor model indicating that the relationship between variables in the data set is mostly linear. As per the results obtained, the real estate business should pay attention to housing having larger area, number of bathrooms, availability of parking spaces, and air conditioning.